

PAPER ONLY

Alpaca MCP Server v2

Alpaca AI Trading Agents 2026

EVIDENCE-GATED AUTONOMOUS TRADING

EdgeStack

Every rule survived an attempt to **kill it**.

An agent whose strategy was built backward: months spent trying to disprove candidate edges — and it only trades what survived.

33 years

OF VALIDATION DATA

3

BACKTEST ENGINES FORCED TO
AGREE

110

RESEARCH SCRIPTS, ALL PUBLIC

1

GRAVEYARD OF REJECTED IDEAS

LLM trading agents lose money **confidently** — their rules come from vibes.

Ask a model for a strategy and it gives you one: plausible, articulate, untested. The field's fix is to wrap the model in deterministic guardrails — and this event is full of excellent guardrails. But guardrails only decide *whether a proposal is allowed*. Nobody asks the harder question:

What evidence does a trading rule need before it deserves to exist?

The graveyard is the point.

- **Elliott waves** — shuffled-return surrogates reproduce the "patterns" (corr -0.525 vs -0.526)
- **Fibonacci retracements** — 38.2/50/61.8 rank 4th–14th of 28 arbitrary bands
- **Intraday mean reversion** — the "edge" was bid-ask bounce (trade AC -0.43 , midquote $+0.07$)
- **Leveraged-ETF decay shorting** — the drag is real ($t = -5.4$) and drift swamps it 10:1
- **Five macro overlays** — four contradicted out-of-sample, including one we had already explained
- **Our own first options design** — negative expectancy from quoting the worst fill. Found, measured, fixed.

Every rejection is documented with the number that killed it. The negative results are load-bearing: they are why the survivors can be trusted.

Three components. Every number is a measurement.

COMPONENT	RULE	EVIDENCE
Overnight core	Long SPY close→open only, gated by 12-month trend AND the credit canary	Sharpe 0.89 overnight vs 0.05 intraday · 8/9 eras since 1993
Capitulation sleeve	Buy 5σ panics on heavy-but-not-extreme volume (1.4–2.5×), 7 ETFs, 3-session hold	+1.42%/event · 67.6% win · t=4.27 · 136 events / 33y · surrogate-tested
Options component	Defined-risk bull put spreads behind 14 deterministic gates	direction validated in the underlying; sized as a satellite — we measured why (spreads eat retail edge)

The index earns its return while the market is closed.



Sharpe ratio, 1993–2026. $\text{corr}(\text{overnight}, \text{intraday}) = 0.004$ — dropping the intraday half is risk reduction, not relabeling.

$t(\text{diff})=6.58$, out-of-sample, $n=4,359$. The same boundary appears in the volume ceiling and the credit canary: **markets revert emotional moves and honor informational ones.**

Every parameter tuning **failed validation**. The untuned rules won — and that is the point.

CONFIG	TRAIN SHARPE (2008–17)	VALIDATION SHARPE (2018–26)	VALIDATION DD
always-on core (no gates)	0.51 · DD –50.7%	—	—
research defaults	0.80	0.65	21.2%
"improved" (tuned on train)	0.89	0.44	28.1%
+ credit canary (from the trader's own older strategy — adopted only after passing BOTH windows)	0.98	1.02	11.8%

A rule change counts only if it survives a disjoint window. One did. Full protocol: ENGINE-TRIAL.md.

Signals are arithmetic. Gates are pure functions. The LLM narrates — it never sizes, prices, or authorizes.

- **Production engine reproduces the 33-year research record to 3 decimals** — research and live code cannot drift apart silently (test_signal_engine.py)
- **14 deterministic risk gates**, 22-case suite where the *right* gate must object — a gate that never fires is not a control
- **Alpaca MCP Server v2 integration is real**: account reads and order submission route through MCP (streamable-http), REST fallback logged per call
- **Append-only decision journal** — every session recorded, including refusals, with the evidence for each
- **Self-hosted, unattended**: ensure-running supervisors, logon persistence, public live dashboard

EdgeStack • LIVE

Every rule survived an attempt to kill it · 33 years of evidence, three backtest engines, one graveyard · Alpaca paper account PA3ZCDD0PR2N

EQUITY

\$100,000

+0 vs \$100k start

EQUITY GATE

CLOSED

trend UP (12m +19.0%) | credit
DETERIORATING (HYG 79.74 vs SMA100
79.85)

MACRO REGIME

TLT 21d sd 0.555 vs 90d mean 0.760
(ratio 0.73) -> CALM

ALPACA MCP SERVER

ROUTING

orders + account via MCP, REST
fallback

OPEN POSITIONS

SYMBOL	COMPONENT	SIZE	ENTERED
flat — waiting for signals that clear the gates			

STRATEGY — EVERY NUMBER IS A MEASUREMENT

- SPY overnight core · Sharpe 0.89 vs 0.05 intraday · 8/9 eras
- gate: 12-month trend AND credit canary (HYG > SMA100) · validated on disjoint windows 0.80→0.98 / 0.65→1.02
- capitulation sleeve · +1.42%/event · 67.6% win · t=4.27 · 33y
- volume ceiling 2.5x — above it "real news arrived", edge dies
- options: defined-risk put spreads behind 14 deterministic gates

DECISION JOURNAL (LATEST SESSIONS)

SESSION	SIGNALS	WHAT HAPPENED & WHY
2026-08-30	0	core_gated: trend UP (12m +19.0%) credit DETERIORATING (HYG 79.74 vs SMA100 79.8
2026-08-30	0	no trade — gates held
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The equity gate is **CLOSED**: trend is up, but credit slipped below its 100-day average — risk is being repriced on information, so the agent refuses. **Refusal is a decision with evidence attached.**

What we will not pretend.

- The equity edges are **risk-adjusted** edges. A 5-session P&L window is mostly noise, for every entrant — we say so out loud.
- Signals are rare by design. The gates refuse most sessions. **Flat is a position.**
- Option-level expectancy could not be established from free historical data (no quote history exists on this tier) — so the options book is a satellite, priced from live quotes only, and the write-up explains exactly why.

An entry that hides its limits is asking judges to find them. We measured ours first.

Rules that earned their existence.
An engine that provably implements them.
An agent that explains every refusal.

github.com/jpennin5/edgestack · live dashboard (URL in repo) · Alpaca paper PA3ZCDD0PR2N · demo video:
[docs/demo.mp4](#)

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